Stabilised GRPO (ours)
Standard GRPO
SFT Reader (start of GRPO)

Training reward ↑
1.0
0.8
0.6
0
10
20
30
40
50
60
70
GRPO step

(a)

Policy entropy
2.0
1.5
1.0
0.5
best checkpoint of
standard GRPO
0
10
20
30
40
50
60
70
GRPO step

(b)

KL to reference policy
0.15
0.10
0.05
0.00
0
10
20
30
40
50
60
70
GRPO step

(c)

Dev30 accuracy (%) ↑
92.5
90.0
87.5
85.0
0
10
20
30
40
50
60
70
GRPO step

(d)